



MATHEMATICS

Rates and Percent Small-Group Lab

Grade 6 | Small-Group Work

Use tables, unit rates, and percentages to compare situations and justify decisions.

INSIDE THIS RESOURCE

Teacher guidance • 16 practice prompts • Reflection exit ticket • Complete answer guidance

Print the practice pages separately from the teacher key. Provide extra paper for extended work.



Teacher guide

NC.6.RP.3 - Use ratio and rate reasoning to solve real-world and mathematical problems, including unit rates and percentages.

Suggested time: 45-55 minutes

Learning focus: Use tables, unit rates, and percentages to compare situations and justify decisions.

Teaching sequence: Review prior instruction and model the first prompt with its answer. Assign one four-prompt page at a time. Ask for a check, a reason, or a specific example. Use the reflection ticket to plan the next lesson; it is not a standardized mastery score.

Model prompt: 3 notebooks cost \$7.50. Unit price?

Model answer: \$2.50

Support options

AIG: justify a second method or create a transfer example. 504: follow the learner's plan for time, breaks, format, and reduced copying. EC: model steps and accept aligned oral, drawn, or written responses. ESL: preview vocabulary, use bilingual references, and rehearse before writing.

Source and scope: North Carolina Department of Public Instruction, Mathematics Standards Vertical Progression: <https://www.dpi.nc.gov/documents/publications/catalog/ma196-vertical-progression-math/open>; NC Check-Ins 2.0 Mathematics Specifications:

<https://www.dpi.nc.gov/documents/accountability/testing/nc-checkin/nc-check-ins-20-mathematics-grades-3-8-specifications> (accessed September 14, 2026).



Practice 1–4

Show work or explain your response. Use additional paper when needed. Teachers may cut apart the four bordered prompts.

1. 3 notebooks cost \$7.50. Unit price?

2. A car travels 180 miles in 3 hours. Unit rate?

3. Complete equivalent ratio $4:7 = 12: \underline{\hspace{1cm}}$.

4. 25% of 80



Practice 5–8

Show work or explain your response. Use additional paper when needed. Teachers may cut apart the four bordered prompts.

5. A \$40 item is discounted 15%. Discount amount?

6. A recipe uses 2 cups for 5 servings. Cups per serving?

7. Which is cheaper: 6 pens for \$9 or 10 for \$16?

8. 18 is what percent of 60?



Practice 9–12

Show work or explain your response. Use additional paper when needed. Teachers may cut apart the four bordered prompts.

9. Increase 50 by 12%.

10. A map scale is 1 inch:8 miles. 3.5 inches represents?

11. A runner completes 5 km in 25 min. Minutes per km?

12. Explain why 0.35 and 35% are equivalent.



Practice 13–16

Show work or explain your response. Use additional paper when needed. Teachers may cut apart the four bordered prompts.

13. Create a ratio table for 4 tickets costing \$30.

14. A class of 24 has 18 present. Percent present?

15. Tax is 7% on \$60. Total?

16. Compare $\frac{7}{10}$, 68%, and 0.72.



Reflection exit ticket

Complete independently. Refer to your completed practice, but explain in your own words. This checks reflection and application, not a test score prediction.

1. State the focus of this resource in your own words.
2. Create one accurate example connected to the practice.
3. Explain how you checked that example.
4. Identify one possible error and show a correction.
5. Name one specific next practice step and explain why it will help.



Teacher answer guidance

1. \$2.50
2. 60 mph
3. 21
4. 20
5. \$6
6. 0.4 cup
7. 6 for \$9; \$1.50 each vs. \$1.60
8. 30%
9. 56
10. 28 miles
11. 5
12. Percent means per hundred; $35/100 = 0.35$
13. 1:\$7.50, 2:\$15, 4:\$30
14. 75%
15. \$64.20
16. $68\% < 7/10 < 0.72$

Reflection criteria

Items 1–4: 2 points for an accurate, specific response; 1 for a partly correct response; 0 for a missing or unrelated response. Accept justified alternatives. Item 5 is an unscored planning reflection. Feedback should identify a next teaching move, not label a learner.

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